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ICICI Bank: AI-Driven Dashboard for Customer Financial Health & Risk Analysis

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Project Domain

Excel / Power BI with Copilot

A Collaborative Initiative

equipping students with practical skills in AI, Data Analytics, and Business Intelligence

by:

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3. Problem Statement

Banks like ICICI generate and collect massive volumes of customer data daily. However, this data is often underutilized due to being spread across multiple systems and lacking a centralized, analytical framework. Managers face challenges such as:

- **Inactive Accounts**: Accounts with little or no activity reduce engagement and revenue.
- **High-Risk Customers / Loan Defaults:** Difficulty in detecting potential loan defaulters or risky financial behavior.
- **Loan Exposure Monitoring:** Tracking total loans per customer segment to prevent overexposure.
- **Customer Segmentation:** Lack of personalized insights limits targeted services and offers.
- **Financial Health Trends:** Monitoring account balances and transactions over time for proactive decisions.
- **Unusual Activity Detection:** Detection of unusual financial patterns based on transaction activity and balance trends.

Business Problem:

ICICI Bank requires an AI-assisted analytical solution to monitor customer financial health, identify inactive and high-risk accounts, track loan exposure, detect unusual behavior, and provide actionable insights for data-driven decision-making.

This project uses a simulated dataset of 1600 ICICI customer records to analyze financial behavior, risk categories, and account activity, transforming raw data into a centralized AI-driven dashboard to support better banking decisions.

4. Objectives

Before diving into the objectives, this project focuses on transforming raw banking data into actionable insights and enhancing financial decision-making using AI-powered analysis and interactive dashboards. The specific objectives of this project are:

Data Exploration & Understanding

- Examine the banking customer dataset to understand its structure, fields, and data quality.
- Identify key attributes related to account activity, loans, risk categories, and customer segments.

Data Cleaning & Preprocessing

- Remove duplicates, handle missing values, and standardize formats for accuracy.
- Ensure the dataset is reliable and ready for analysis and visualization.

Financial Analysis & Risk Assessment

- Analyze customer financial activity, loan exposure, and risk levels using Excel formulas and Power BI measures.
- Detect inactive accounts and identify high-risk customers for proactive management.
- Define logical rules to classify inactive accounts and high-risk customers based on transaction and loan data.

Dashboard Development & Visualization

- Build an interactive AI-driven dashboard displaying total balances, loan exposure, risk categories, and customer segments.
- Enable managers to quickly interpret data through KPI cards, charts, and visual summaries.

AI Integration for Insights

- Use Microsoft Copilot and AI chatbots to generate automated summaries and insights.
- Provide actionable suggestions for engagement, retention, and risk mitigation.

Actionable Insights for Decision-Making

- Transform raw data into meaningful business intelligence for management decisions.
- Support proactive strategies to monitor customer activity and financial health trends.

Documentation & Knowledge Sharing

- Document the entire project workflow and analysis process.
- Create a GitHub repository containing dashboards, dataset, and a demonstration video for easy sharing and evaluation.

5. Methodology / Implementation Plan

This project will be carried out using a structured five-phase workflow to ensure effective data analysis, AI integration, and actionable insights:

#	Phase	Description
1	**Data Collection & Exploration**	Obtain the banking dataset (~1600 records), understand attributes, data types, and identify potential quality issues.
2	**Data Cleaning & Preprocessing**	Remove duplicates, handle missing values, correct formats, and standardize data using Excel formulas and tools (IF, IFERROR, Text-to-Columns).
3	**Analysis & AI Integration**	Analyze trends and patterns using Excel formulas and pivot tables. Apply business rules to identify inactive accounts and high-risk customers using calculated columns. AI tools (Copilot, Chatbots) generate insights and suggestions.
4	**Visualization / Output**	Build interactive charts, KPI cards, and dashboards in Excel or Power BI showing total balances, loan exposure, customer segments, risk categories, and account activity.
5	**Documentation & Submission**	Prepare project report, upload dataset and dashboards to GitHub, and record a video demonstrating project features and insights.

6. Resources and Timeline

A. Hardware & Software Requirements

- Laptop or PC — Intel Core i3 or higher, minimum 4 GB RAM (8 GB recommended)
- Operating System: Windows 10 / 11 or macOS
- Microsoft Excel (Microsoft 365 / Excel 2019 or later)
- Power BI Desktop — latest version (free download from Microsoft) — where applicable

- Microsoft Copilot — via Microsoft 365 subscription — where applicable
- AI Chatbot (ChatGPT / Google Gemini / Microsoft Copilot Web) — for prompt-based projects
- Internet connection — required for Copilot and AI Chatbot access
- Web browser — Google Chrome or Microsoft Edge (latest version)

B. Dataset

The project uses a pre-existing simulated dataset containing 1600 ICICI Bank customer records designed for educational and analytical purposes.

Key Features of the Dataset:

- Customer Name
- Account Type
- Account Balance
- Transaction Count
- Loan Amount
- Loan Status
- Account Status
- Customer Segment
- Risk Category

Purpose and Usage:

- The dataset simulates real-world banking customer information while maintaining data privacy, as actual banking data is confidential.
- It supports data analysis, AI integration, and dashboard development using Microsoft Excel or Power BI with Copilot.
- Microsoft Copilot assists in generating formulas, validating calculations, identifying trends, creating KPIs, and suggesting visualizations.
- Structured AI prompts are used to extract insights, analyze financial trends, and identify high-risk or inactive customers efficiently.

- Business rules are applied to define inactive accounts and high-risk customers, as the dataset does not explicitly contain these labels.

C. Project Timeline

A simple table showing planned tasks and expected completion dates.

Phase	Task	Tentative Completion Date
1	Dataset collection, exploration, and understanding	*Week 1*
2	Data cleaning, preprocessing, and validation / Prompt design	*Week 2*
3	Analysis, AI tool integration (Copilot / Chatbot), and insight generation	*Week 2 – 3*
4	Dashboard / output build, testing, and review	*Week 3*
5	Report writing, GitHub upload, and video recording and submission	*Week 4*

7. Expected Outcomes / Conclusion

By the end of this project, the following outcomes are anticipated:

- **Cleaned and Validated Dataset:** The banking customer dataset will be thoroughly cleaned, preprocessed, and validated, with all steps documented and before-and-after comparisons provided.
- **Interactive AI-Driven Dashboard:** A dynamic dashboard in Excel or Power BI will visualize key financial indicators such as total account balances, loan exposure per segment, active vs. inactive accounts, customer segments, and risk categories.
- **AI-Assisted Insights:** Demonstrated use of Microsoft Copilot and AI chatbots to generate summaries, identify high-risk or inactive accounts based on defined rules, and provide actionable insights for management decisions.
- **Comprehensive Project Repository**: A GitHub repository will house all project files, including datasets, dashboards, reports, and the structured workflow.
- **Demonstration Video:** A recorded walkthrough will showcase the project workflow, dashboard features, KPIs, AI-generated insights, and decision-making support.
- **Practical Application:** The dashboard will enable banking managers to quickly identify inactive accounts, high-risk customers, and loan distribution patterns, facilitating better financial monitoring, risk management, and data-driven strategic decisions.

- **Impact:** This project demonstrates the power of combining data analytics, business intelligence tools, and AI technologies to convert raw banking data into actionable insights that enhance financial health monitoring and operational efficiency.

8. References

List all references, tutorials, and documentation used in the project. Students must cite any material they referred to, including research papers, GitHub repositories, YouTube tutorials, blogs, and official documentation.

- Edunet Foundation. AICW Fellowship Program Guidelines. 2026. Available at: https://www.edunetfoundation.org
- LinkedIn Learning. AI and Data Analytics Courses. 2026. Available at: https://www.linkedin.com/learning
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